

Taeseong Kim

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OVERVIEW

I am an Integrated Ph.D. student at the Department of Mathematical Sciences, Seoul National University (SNU), Republic of Korea. My advisor is Prof. Jung Hee Cheon. I work on cryptography, currently focusing on Homomorphic Encryption (HE) and its applications, including privacy-preserving machine learning.

EDUCATION

Seoul National University, Seoul, Republic of Korea

- Integrated Ph.D. in Mathematical Sciences Sep 2021 – Present
 - Consists of a two-year M.S. course and a three-year Ph.D. course
 - Adviser: Jung Hee Cheon from Mar 2023
 - Focus: Cryptography (Homomorphic Encryption, Privacy-Preserving Machine Learning)

Pusan National University, Busan, Republic of Korea

- B.S. in Mathematical Sciences, Summa Cum Laude Mar 2015 – Aug 2021

PUBLICATIONS

Authors are listed in alphabetical order by last name, unless an asterisk(*) is indicated.

CONFERENCES

- C01 Wonhee Cho, Guillaume Hanrot, Taeseong Kim, Minje Park, and Damien Stehlé, “Fast and Accurate Homomorphic Softmax Evaluation,” *ACM SIGSAC CCS 2024*.
- C02 *Dongwhan Rho[†], Taeseong Kim[†], Minje Park, Jung Woo Kim, Hyunsik Chae, Jung Hee Cheon, Ernest K. Ryu, “Encryption-Friendly LLM Architecture,” *ICLR 2025*,

[†] denotes first co-first author

- C03 Youngjin Bae, Jung Hee Cheon, Minsik Kang, Taeseong Kim[†], “High-Throughput AES Transciphering using CKKS: Less than 1ms,” *WAHC 2025*

[†] corresponding author

JOURNALS

- J01 Jung Hee Cheon, Minsik Kang, Taeseong Kim, Junyoung Jung, and Yongdong Yeo, “Batch Inference on Deep Convolutional Neural Networks With Fully Homomorphic Encryption Using Channel-By-Channel Convolutions,” *IEEE Transactions on Dependable and Secure Computing*, Aug 2024.

MANUSCRIPTS

- M01 Jung Hee Cheon, Seungwan Hong, Minsik Kang, Jonghyun Kim, Taeseong Kim, Changmin Lee, Junho Lee “MRFHE: Mixed-Radix Fully Homomorphic Encryption with Better Batch Bootstrapping,” *eprint/2026/853*

AWARDS & HONORS

AWARDS

- Excellent Prize (3rd place) Oct 2024
(Korean) National Cryptography Contest, National Security Research Institute (NSRI)
for the conference paper C01: “Fast and Accurate Homomorphic Softmax Evaluation”
- Special Prize Oct 2023
(Korean) National Cryptography Contest, National Security Research Institute (NSRI)
for the journal paper J01: “Batch Inference on Deep Convolutional Neural Networks With Fully Homomorphic Encryption Using Channel-By-Channel Convolutions”
- First Place Scholarship, Software-Focused University Program in undergraduate Dec 2019

HONORS

- SNU Teaching and Research Support Scholarship Sep 2025 – Feb 2026
- BK 21+ Scholarship Aug 2022 – Feb 2023, Feb 2024 – Aug 2024
Ministry of Education of Korea
- National Scholarship for Outstanding Students in Science and Engineering Sep 2018 – Feb 2021
Korea Student Aid Foundation

TALKS

2025

- High-Throughput AES Transciphering using CKKS: Less than 1ms Apr 2025

2025 Korea Mathematics Society Spring Meeting, Daejeon, South Korea
ACM SIGSAC CCS 2025 Workshop WAHC, Taipei, Taiwan

2024

- Encryption-Friendly LLM Architecture Oct 2024
2024 Korea Mathematics Society Fall Meeting, Suwon, South Korea
- Fast and Accurate Homomorphic Softmax Evaluation Oct 2024 May 2024
ACS SIGSAC CCS 2024, Salt Lake City, USA
2024 Korea Mathematics Society Spring Meeting, Daejeon, South Korea

2023

- Batch Inference on Deep Convolutional Neural Networks With Fully Homomorphic Encryption Using Channel-By-Channel Convolutions Oct 2023
2023 Korea Mathematics Society Fall Meeting, Seoul, South Korea

PROJECTS

List of selective projects.

- DARPA Data Protection in Virtual Environments (DPRIVE) 2023
- Development and Library Implementation of Fully Homomorphic ML Algorithms supporting Neural Network Learning over Encrypted Data (IITP) 2023 – Aug 2024
- Sensitive Data Protection using HE and its Acceleration (Samsung Elec.) 2023 – Aug 2025

EXPERIENCES

TEACHING

- Seoul National University, Math Courses TA
 - Introduction to Cryptography Sep 2025 – Feb 2026
 - Calculus for Humanities and Social Sciences 1 Sep 2024 – Feb 2025
 - Calculus Practice 2 Sep 2023 – Feb 2024 , Sep 2021 – Feb 2022
 - Calculus Practice 1 Mar 2023 – Aug 2023, Mar 2025
 - Differential Equation and Practice Sep 2022 – Feb 2023
 - Linear Algebra Mar 2022 – Aug 2022

INTERNSHIPS

- Research Internships Sep 2025 – Feb 2026
Research intership (part-time) at CryptoLab
- Research Internships Jul 2025 – Sep 2025
Research intership (full-time) at CryptoLab and FHELab in France
- Undergraduate Internships
 - Canada Internship Program Jan 2019 – Mar 2019
Developed in-house applications for controlling the flow of stocks.

EDUCATION PROGRAM

- Undergraduate international education program
 - SW Pioneer Program in the University of Southern California (USC) Jul 2019 – Aug 2019

SKILLS

- $\text{\LaTeX}$, Matlab, Python: Proficient
- C/C++, SageMath, Sollya: Working Knowledge
- R, PyTorch, TensorFlow: Basic

SERVICES

REVIEWER (CONFERENCES)

- WHAC 2023, PKC 2024, Eurocrypt 2024, PQCrypto 2024, Asiacrypt 2025, IEEE SP 2026.

Last Updated: Mar 2026